

Exploring new physics with proton interactions at the LHC

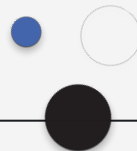


LABORATÓRIO DE INSTRUMENTAÇÃO
E FÍSICA EXPERIMENTAL DE PARTÍCULAS

Physics of anomalous tau couplings

Recent research has focused on lepton pair, WW/ZZ boson and top quark ($t\bar{t}$) production, leading to:

- Precision Tests of the Standard Model
- Enhanced Sensitivity to New Physics: the ability to isolate exclusive events makes CEPs valuable for searching BSM;
- Forward Proton Tagging: the use of forward proton detectors – like PPS – improves event selection and provides more information about scattered protons, leading to more accurate measurements;



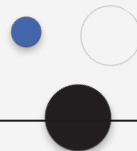
Analysis

- Proton collisions at the LHC offer a unique environment for probing the Standard Model (SM)
- These collisions allow exploration of rare processes and potential new physics that are not accessible in lower-energy experiments.

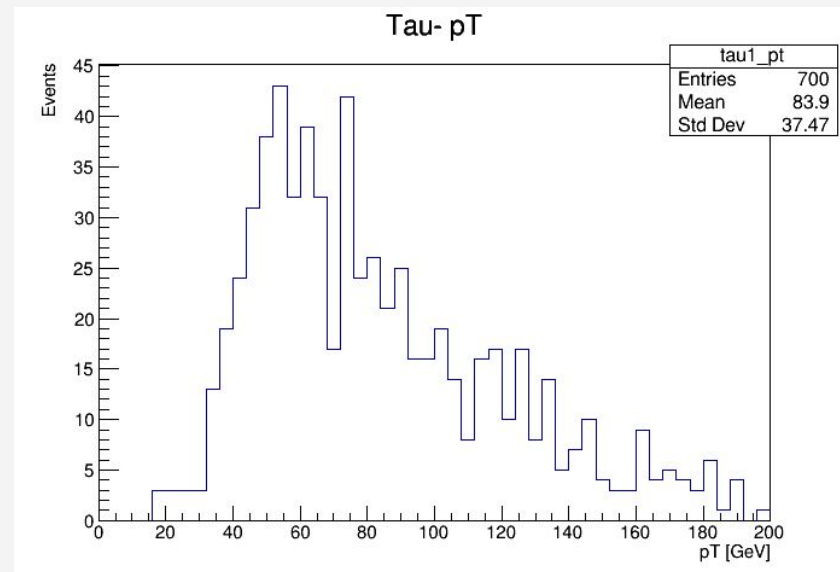
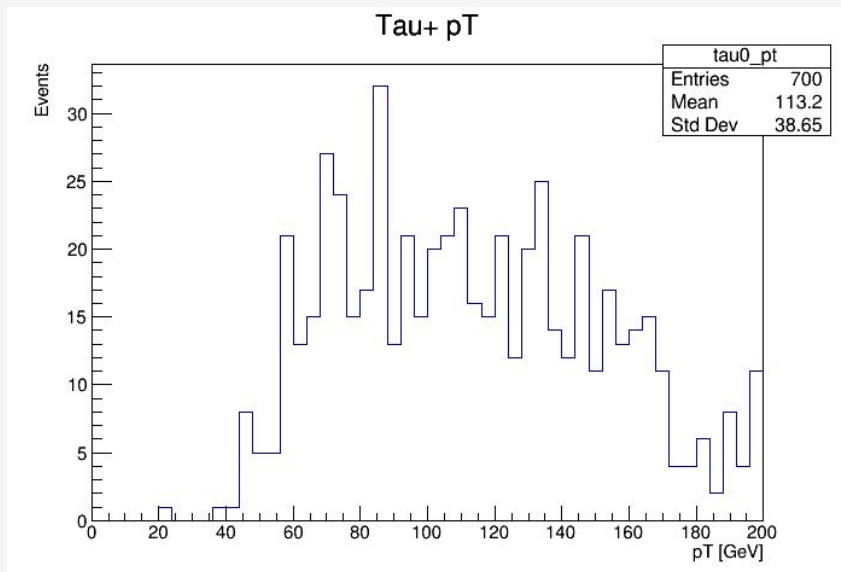


Analysis Improvements

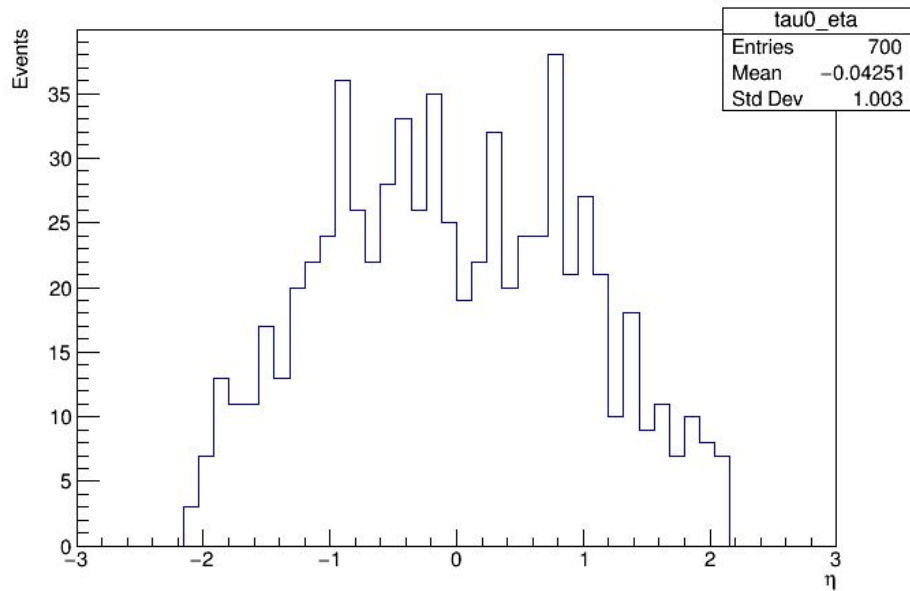
- Exploring the Anomalous Magnetic Moment of the Tau Lepton: unlike electrons and muons, leptons have higher mass and shorter lifetime, making it challenging to study the tau's $g-2$ - requires high-energy experiments;
- Since taus are 3rd generation of leptons and heavier than electrons and muons, they may be even more sensitive to Beyond SM physics;



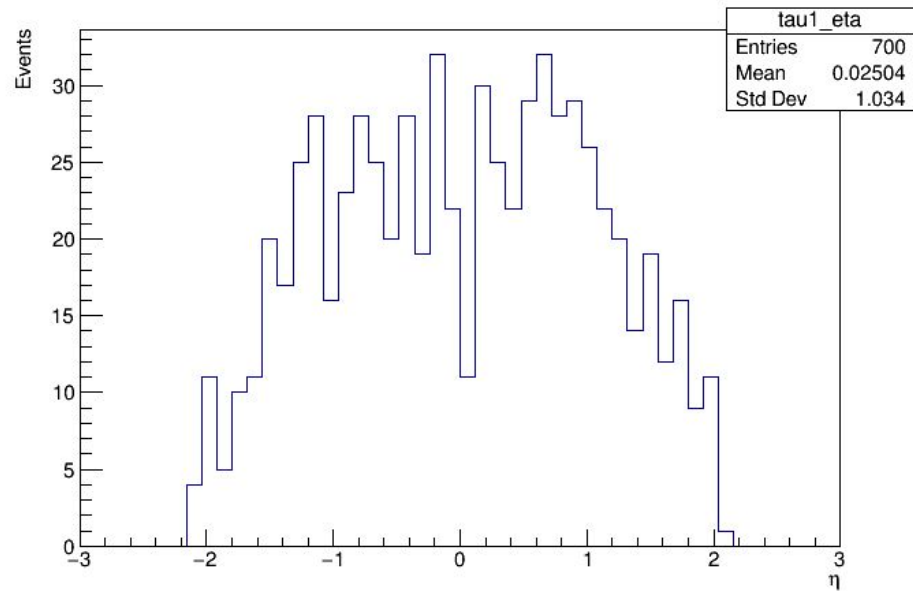
few plots – show properties of expected signal



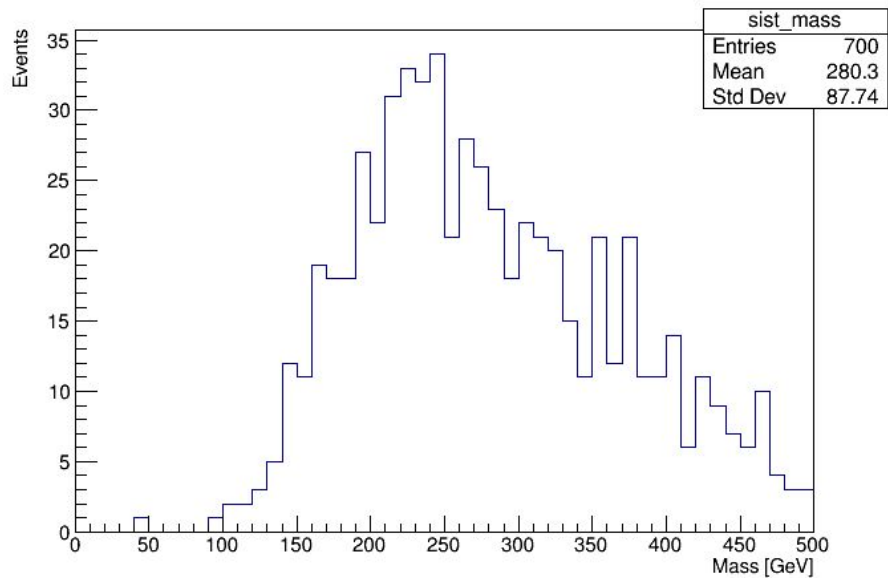
Tau+ Eta



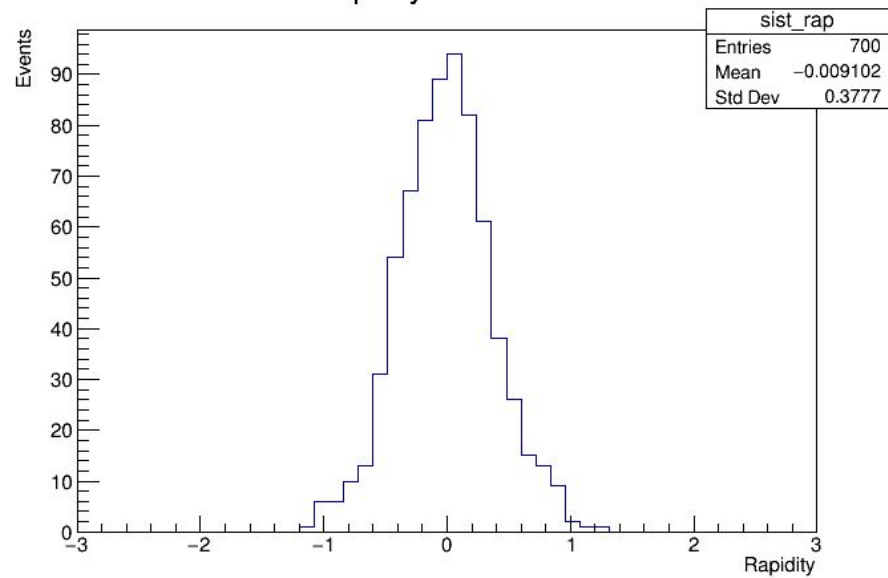
Tau- Eta



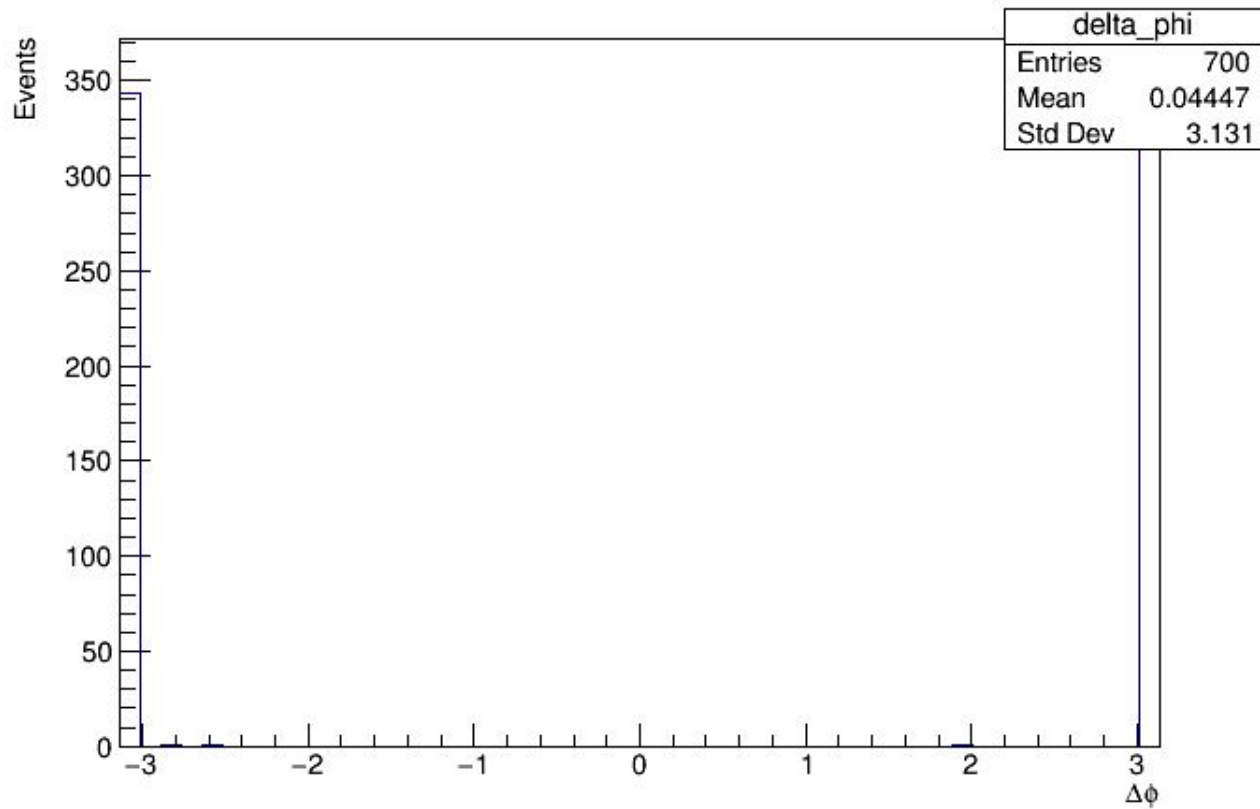
Invariant Mass of Tau Pair



Rapidity of Tau Pair

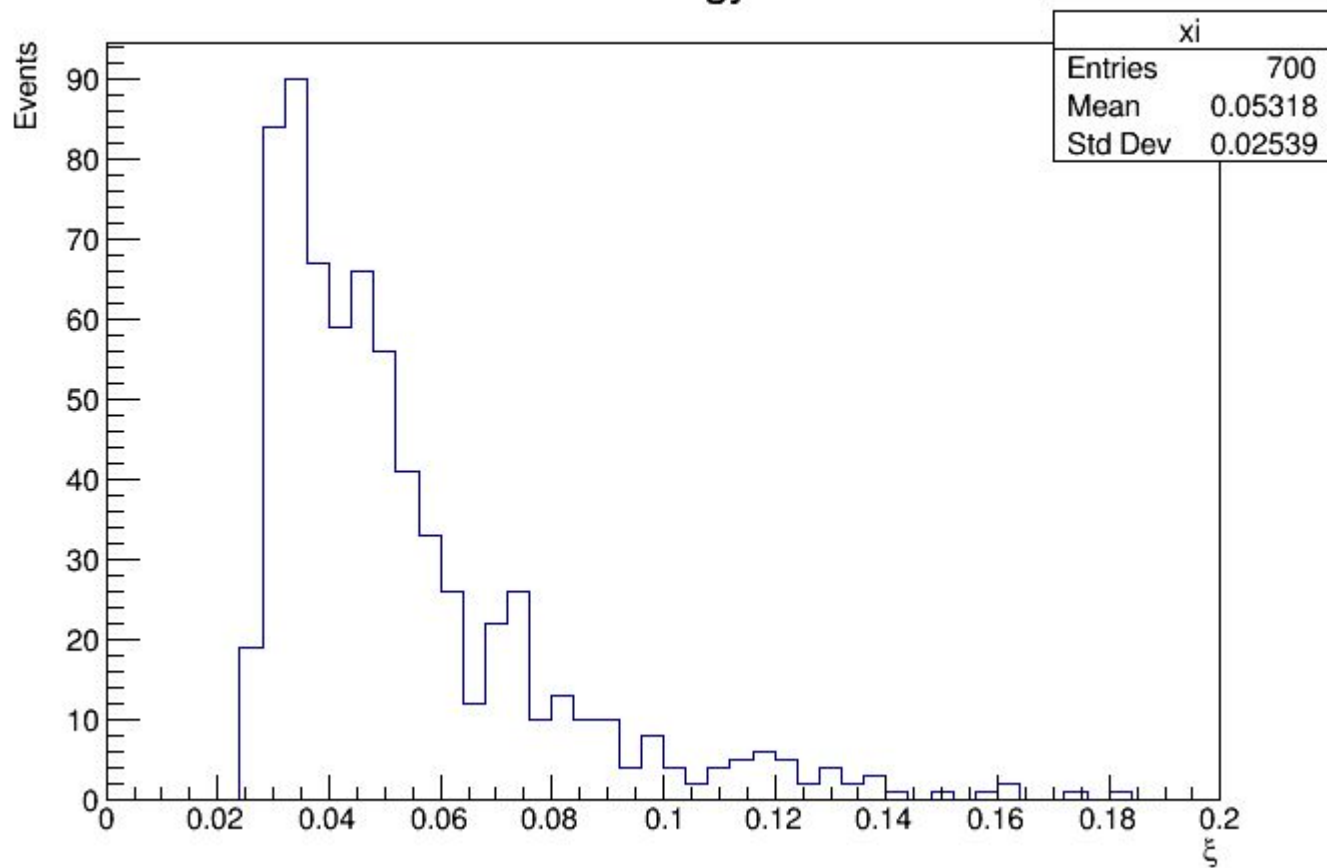


Delta Phi Between Tau+ and Tau-



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Successfully loaded TTree: tree  
tau0_phi: 6.93062e-310, tau1_phi: 0
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Proton Energy Loss



Thank you
for your attention